Particle Flow Filter and Differentiable Particle Filter

Report Structure

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Section A. A literature review

Modern financial systems generate large volumes of sequential data, including asset prices, volatility indicators, and transaction records. Extracting the underlying latent dynamics of such systems is central to risk management, portfolio optimization, and anomaly detection. State Space Models (SSMs) provide a principled probabilistic framework for representing these temporal processes.

In the special case where both the state transition and observation models are linear and Gaussian, the Kalman Filter (KF) provides an optimal minimum-variance estimator (Pei et al., 2019; kal, 2025). Its closed-form prediction and update equations yield an efficient recursive solution, making it a cornerstone of linear state estimation. Nevertheless, its optimality depends critically on the linear–Gaussian assumptions. Moreover, even within this setting, numerical challenges may arise in practical implementations: the covariance matrix can lose symmetry or positive definiteness due to finite-precision arithmetic. To mitigate such issues, the Joseph-stabilized covariance update is commonly employed to enhance numerical robustness (kal, 2025). Recent research has further emphasized the importance of formally verifying the reliability of Kalman filter implementations, for example through probabilistic model checking techniques (Evangelidis and Parker, 2019).

Moving beyond the linear–Gaussian setting considered above, real-world financial systems are frequently characterized by pronounced nonlinearities, posing significant challenges for classical state-space estimation methods. When the state transition or observation equations are nonlinear, the exact Kalman filtering recursion is no longer analytically tractable, as the posterior distribution cannot be expressed in closed form. To address this limitation while preserving the recursive structure of the Kalman framework, the Extended Kalman Filter (EKF) introduces a local linear approximation of the nonlinear system around the current state estimate (Pei et al., 2019). By applying first-order Taylor expansions, the EKF enables approximate Gaussian filtering in nonlinear settings. However, the accuracy of this approach depends critically on the validity of the linear approximation. If the underlying dynamics exhibit strong curvature or if the linearization is performed far from the true state, approximation errors may accumulate, potentially resulting in biased estimates or filter divergence.

To improve the treatment of nonlinear transformations, the Unscented Kalman Filter (UKF) replaces analytic linearization with a deterministic sampling strategy. Rather than approximating the nonlinear function directly, the UKF propagates a carefully selected set of sigma points through the nonlinear transformation and reconstructs the posterior mean and covariance from the transformed points (Pei et al., 2019; Wan and Van Der Merwe, 2000). This unscented transform typically provides a more accurate approximation of moment propagation without requiring explicit Jacobian calculations, which are necessary in the EKF for first-order Taylor linearization. Nevertheless, despite this improvement, both EKF and UKF rely on approximate representations of nonlinear transformations. In systems characterized by severe nonlinearities—commonly observed in real-world financial environments—these

approximations may become inaccurate, leading to degraded filtering performance.

When nonlinear effects cause the filtering distribution to deviate substantially from Gaussianity or to exhibit complex structure, a more flexible representation may be required. Sequential Monte Carlo (SMC) methods, commonly referred to as Particle Filters (PFs) in the context of state-space models, provide such an alternative (Doucet and Johansen, 2009). Unlike EKF and UKF, which approximate the filtering distribution by a Gaussian characterized solely by its mean and covariance, PFs approximate the entire distribution using a collection of weighted samples, or particles.

At each time step, particles are propagated through the state transition model (prediction step), reweighted according to the likelihood of the new observation (update step), and optionally resampled to focus computational effort on high-probability regions. This sampling-based representation enables inference in arbitrary nonlinear and non-Gaussian state-space models and, in principle, allows accurate estimation even when the filtering distribution is heavy-tailed or multimodal—features commonly encountered in complex financial systems. Algorithm 1 shows a standard particle filter. For more details regarding the formulas, please see Corenflos et al. (2021).

Algorithm 1 Standard Particle Filter (Corenflos et al., 2021)

Sample $X_1^i \overset{\text{i.i.d.}}{\sim} q_\phi(\cdot \mid y_1)$ for $i \in [N]$
Compute $\omega_1^i = \frac{p_\theta(X_1^i, y_1)}{q_\phi(X_1^i \mid y_1)}$ for $i \in [N]$
$\hat{\ell}(\theta) \leftarrow \frac{1}{N} \sum_{i=1}^N \omega_1^i$
for $t = 2, \ldots, T$ **do**
    Normalize weights $w_{t-1}^i \propto \omega_{t-1}^i, \quad \sum_{i=1}^N w_{t-1}^i = 1$
    Resample $\tilde{X}_{t-1}^i \sim \sum_{i=1}^N w_{t-1}^i \delta_{X_{t-1}^i}$ for $i \in [N]$
    Sample $X_t^i \sim q_\phi(\cdot \mid \tilde{X}_{t-1}^i, y_t)$ for $i \in [N]$
    Compute

$$\omega_t^i = \frac{p_\theta(X_t^i, y_t \mid \tilde{X}_{t-1}^i)}{q_\phi(X_t^i \mid \tilde{X}_{t-1}^i, y_t)} = \frac{p_\theta(y_t \mid X_t^i)\, p_\theta(X_t^i \mid \tilde{X}_{t-1}^i)}{q_\phi(X_t^i \mid \tilde{X}_{t-1}^i, y_t)} \quad \text{for } i \in [N]$$

    Compute $\hat{p}_\theta(y_t \mid y_{1:t-1}) = \frac{1}{N} \sum_{i=1}^N \omega_t^i$
    $\hat{\ell}(\theta) \leftarrow \hat{\ell}(\theta) + \log \hat{p}_\theta(y_t \mid y_{1:t-1})$
return log-likelihood estimate: $\hat{\ell}(\theta) = \log \hat{p}_\theta(y_{1:T})$

In practice, however, PFs are well known to suffer from degeneracy (Doucet and Johansen, 2009): after a few iterations, most particles carry negligible weight, with only a small fraction effectively representing the posterior. Resampling mitigates weight collapse at the current time step but does not eliminate the fundamental problem.

Daum and Huang (2011) identify the measurement update—the evaluation of the product of the prior and likelihood at particle locations—as the root cause of degeneracy in particle filtering. In a standard particle filter, particles are first propagated to represent the prior distribution and are then reweighted according to the likelihood of the latest observation. When the likelihood is highly concentrated and its peak differs from that of the prior, most particles sampled from the prior fall into regions with very low likelihood and therefore receive negligible weights. As a result, the effective sample size collapses and the posterior approximation becomes dominated by only a few particles. From this perspective, degeneracy is not merely a numerical issue but a structural consequence of implementing Bayes' rule through pointwise multiplication at fixed particle locations.

To address this issue, Daum et al. (2010); Daum and Huang (2011) propose replacing the discrete Bayesian update with a continuous density transport formulation. They introduce a log-homotopy that defines a smooth family of intermediate densities parameterized by a synthetic variable $\lambda \in [0, 1]$, continuously connecting the prior at $\lambda = 0$ to the posterior at $\lambda = 1$. Particle trajectories are then governed by an ordinary differential equation (ODE) derived from the continuity equation, which ensures that the evolving particle distribution is consistent with this intermediate density. By continuously transporting particles from the prior to the posterior through a flow that preserves probability mass at each intermediate step—without the need for importance weighting or resampling—the method alleviates the degeneracy problem inherent in conventional particle filters and can achieve accurate estimates with significantly fewer particles.

The evolution of the intermediate density $p(x, \lambda)$ induced by the particle flow is governed by a partial differential equation derived from the Fokker–Planck equation Daum and Huang (2011). In the absence of process noise (i.e., zero diffusion), the resulting formulation corresponds to a deterministic particle flow, where particle trajectories follow a deterministic differential equation. In contrast, when non-zero process noise is introduced, the formulation leads to a stochastic particle flow, in which particle trajectories are governed by a stochastic differential equation.

Deterministic particle flow methods transport particles from the prior distribution to the posterior distribution within a single time step. In practice, the state distribution is represented by a set of particles, but the exact analytical form of the underlying distribution is unknown. However, many particle flow formulations require the mean and covariance of the state distribution during the prediction and update steps, since these quantities often appear as parameters in the mathematical formulation of the particle flow. For example, the Exact Flow Daum–Huang (EDH) filter (Ding and Coates, 2012; Daum et al., 2010) requires the prediction error covariance matrix during implementation. Therefore, the covariance must be estimated in order to define the particle flow at each step. The Extended Kalman Filter (EKF) or the Unscented Kalman Filter (UKF) can be used to estimate the required covariance, since it can be easily computed using EKF or UKF methods Daum and Huang (2011). Regarding the algorithmic implementation, Ding and Coates (2012) provides a detailed description of the EDH filter and introduces its locally exact extension (LEDH), motivated by the limitations of global linearization in EDH. In LEDH, the particle flow parameters are

computed using particle-wise linearizations, allowing the flow dynamics to adapt locally to the geometry of the likelihood.

Stochastic particle flow extends deterministic particle flow by introducing a non-zero diffusion term into the particle dynamics, allowing particles to evolve according to a stochastic process during the homotopy transformation. The deterministic formulation can therefore be viewed as a special case of stochastic particle flow in which the diffusion term is zero. Dai and Daum (2021) derive a parameterized family of stochastic particle flows in which the parameterization arises from the introduction of a non-zero diffusion term, which can also improve numerical stability. This formulation provides a unified framework showing that several previously proposed particle flow methods can be interpreted as special cases of this general stochastic flow. For example, the EDH flow can be viewed as a special case within this framework. Furthermore, under Gaussian distribution assumptions, Dai and Daum (2021) demonstrate that particle flows produce unbiased estimates and that the resulting estimators are consistent.

In stochastic or deterministic particle flow filters (PFFs), particles are moved according to a differential equation (Crouse, 2021). The specific equation determines the type of PFF. However, the differential equations governing these flows can be stiff, making their numerical integration challenging (Crouse, 2021). As discussed in Dai and Daum (2022), a particle flow equation is considered stiff when the Jacobian of the flow dynamics has eigenvalues with widely different magnitudes, resulting in a large stiffness ratio and forcing numerical solvers to use very small integration step sizes. Consequently, it can be difficult to determine whether poor performance arises from numerical integration errors or from theoretical limitations of the flow (Crouse, 2021). To address this issue, Dai and Daum (2021) showed that introducing a non-zero diffusion term in stochastic particle flows can improve numerical stability based on the straight-line log-homotopy transformation. Subsequently, Dai and Daum (2022) considered a more general class of λ-varying linear log-homotopy transformations and formulated stiffness mitigation as an optimal control problem to derive an optimal homotopy that mitigates stiffness. Detailed theoretical analysis and proofs are provided in Dai and Daum (2021, 2022).

Another related development in particle flow methods is that the mapping-based particle flow framework, which combines particle filtering with optimal transport through kernel-based mappings in a reproducing kernel Hilbert space (RKHS) (Pulido and van Leeuwen, 2019). By constraining the partcile flow within an RKHS, Pulido and van Leeuwen (2019) derive an analytical expression for the gradient of the Kullback–Leibler (KL) divergence between the intermediate probability density functions (pdfs) and the posterior pdf. This gradient defines a deterministic particle flow that iteratively transports particles from the prior distribution toward the posterior. The approach allows the filter to operate without resampling and maintains equal particle weights. Pulido and van Leeuwen (2019) adopt a scalar Gaussian kernel for relatively low-dimensional systems. With a scalar kernel, however, the kernel value is identical across all components of any two particles, and the divergence of the kernel is proportional to the kernel value itself. Consequently, each component of the kernel's divergence is scaled by the same kernel value Hu and Van Leeuwen (2021). How-

ever, in sparsely observed high-dimensional systems, the gradient of the posterior density can vary substantially across variables. In such cases, the scalar kernel may fail to preserve the variance of the marginal distributions of the observed variables. To address this limitation, a matrix-valued Gaussian kernel is proposed. The proposed kernel is a diagonal matrix whose entries consist of different scalar Gaussian kernels. This formulation allows distances between particles to be measured independently in each dimension, preventing the collapse of marginal distributions Hu and Van Leeuwen (2021).

For all particle flow filters, although the particle flow is formulated as a continuous partial differential equation, practical implementation requires discretizing the flow into finite computational steps. This discretization introduces additional approximation errors during numerical integration (Li et al., 2016). For example, while EDH and LEDH provide effective deterministic mechanisms for transporting particles toward the posterior, discretization and linearization errors imply that the resulting particles are not exact posterior samples. Li and Coates (2017) address this limitation by embedding deterministic particle flows within a standard particle filtering framework, in which the flow defines an invertible deterministic proposal mapping and the associated Jacobian is used to compute importance weights. Importance weighting and resampling are then applied to correct for approximation errors, restoring the theoretical guarantees of sequential Monte Carlo while retaining the robustness of particle flow in high-dimensional and highly informative filtering problems.

So far, we have discussed different frameworks for state-space models. In practice, however, most state-space models depend on parameters. For example, in our algorithm 1, parameters θ and ϕ appear in State 1 and State 2. In the frameworks and methods discussed above, we have assumed that these parameters are known. If the parameters are unknown, they can be estimated from the data.

In many applications, the primary objective is to infer the parameter θ (Kantas et al., 2015). For instance, in biochemical networks the aim is not to estimate the species populations directly, but to identify the chemical reaction rate constants that act as parameters in the transition dynamics. Kantas et al. (2009, 2015) describe two main approaches for estimating such parameters, both in off-line and on-line settings. In the off-line approach, parameters are inferred using a fixed set of observations. In contrast, the on-line approach updates the parameter estimates sequentially as new observations become available Kantas et al. (2015). One approach is the Bayesian framework, where the unknown parameter is treated as a random variable and assigned a prior distribution. The goal is then to determine the posterior distribution of this parameter given the observed data. Another approach is Maximum Likelihood estimation, where the parameter estimate is obtained by maximizing the marginal likelihood of the observed data. More details can be found in Kantas et al. (2009, 2015).

Although parameter estimation for particle filters has been extensively studied, many existing methods assume that the structures of the dynamic (transition model) and measurement (observation model) models, or parts of their parameters, are known (Chen and Li, 2023). However, real-world scenarios often involve high-dimensional and unstructured

observations, such as images. In this context, differentiable particle filters (DPFs) provide a promising approach (Jonschkowski et al., 2018).

Roughly speaking, differentiable particle filters integrate particle filters with deep learning methods. They learn data-adaptive filtering mechanisms by parameterizing particle filter components with neural networks and optimizing them end-to-end via gradient-based learning (Chen and Li, 2023). Chen and Li (2023) emphasise five key design choices in differentiable particle filters, including dynamic models, measurement models, proposal distributions, optimisation objectives, and differentiable resampling techniques. Among these components, the dynamic model, measurement model, and proposal distribution can be implemented using neural networks. In this section, we focus on explaining differentiable resampling techniques. For the remaining components, the reader is referred to Chen and Li (2023) for more details.

As mentioned above, resampling mitigates weight collapse in PFs. However, it is also a major obstacle to achieving fully differentiable particle filters for two reasons: resampling produces discrete outputs and resets particle weights to a constant, making gradients across time steps zero and thus rendering the operation non-differentiable Chen and Li (2023). To address this issue, several differentiable resampling strategies have been proposed. One commonly used approach is soft resampling, which modifies the resampling distribution by interpolating between the original particle weight distribution and a uniform distribution Karkus et al. (2018). Specifically, the modified weight is defined as $\tilde{W}_t^i = \lambda W_t^i + (1-\lambda)\frac{1}{N_p}$, where W_t^i is the original normalized weight, N_p is the number of particles, and $\lambda \in [0,1]$ controls the degree of interpolation. Particles are then sampled from this softened multinomial distribution, and the importance weights are corrected after resampling to account for the modified sampling distribution. This procedure allows the new weights to remain dependent on the original weights, enabling non-zero gradients with respect to pre-resampling weights. Nevertheless, soft resampling still relies on stochastic sampling from a multinomial distribution, meaning that gradients are computed while the inherently non-differentiable component of the resampling step is ignored (Chen and Li, 2023; Corenflos et al., 2021).

An alternative approach formulates the resampling process as an optimal transport (OT) problem, leading to differentiable resampling via entropy-regularised optimal transport (Corenflos et al., 2021). In this framework, resampling is interpreted as transporting particles from an equally weighted empirical distribution to the target weighted particle distribution. Instead of selecting particles through discrete sampling, the method computes an optimal transport plan that redistributes particle mass with minimal cost. The resampled particles are obtained through a barycentric projection, where each new particle is expressed as a weighted combination of the original particles. To make the optimisation computationally tractable and differentiable, an entropy regularisation term is introduced and the resulting problem is solved using the Sinkhorn algorithm. This formulation produces a deterministic and fully differentiable mapping from the original particle set to the resampled particle set, enabling gradient propagation through the resampling step during training. However, the entropy regularisation introduces a small bias, which diminishes as

the regularisation coefficient approaches zero.

Section B. From classical filters to particle flows (Part I)

Implement the Kalman filter for a multidimensional linear-Gaussian SSM. (Do not use `tfp.distributions.LinearGaussianStateSpaceModel`). Use synthetic data from a standard LGSSM, see examples 2 in Pei et al. (2019).

We have Implement the Kalman filter for a multidimensional linear-Gaussian SSM and use sythetic data form staddard LGSSM as examples 2 in Pei et al. (2019). For more details, see `https://github.com/ChunchaoPeter/ml_simulation_research/tree/features_kf_lgssm/kf_lgssm`.

Based on Pei et al. (2019), we consider a discrete-time linear dynamical system of the form:

$$\mathbf{x}_t = F_t \mathbf{x}_{t-1} + B_t \mathbf{u}_t + \mathbf{w}_t, \qquad \mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, Q_t), \tag{1}$$

$$\mathbf{z}_t = H_t \mathbf{x}_t + \mathbf{v}_t, \qquad \mathbf{v}_t \sim \mathcal{N}(\mathbf{0}, R_t), \tag{2}$$

where

- $\mathbf{x}_t \in \mathbb{R}^{n_x}$ is the latent (hidden) state vector,
- $\mathbf{z}_t \in \mathbb{R}^{n_z}$ is the observation vector,
- $\mathbf{u}_t$ is an optional control input,
- $\mathbf{w}_t, \mathbf{v}_t$ are mutually independent Gaussian noise terms,
- F_t, B_t, H_t are known model matrices,
- Q_t and R_t are the process and observation covariance matrices.

The Kalman filter provides recursive minimum-variance estimates of $\mathbf{x}_t$ given the observations $\mathbf{z}_{1:t}$.

Initialization

$$\hat{\mathbf{x}}_{0|0} = \mathbb{E}[\mathbf{x}_0], \qquad \Sigma_{0|0} = \mathrm{Cov}[\mathbf{x}_0]. \tag{3}$$

Prediction Step

$$\hat{\mathbf{x}}_{t|t-1} = F_t \hat{\mathbf{x}}_{t-1|t-1} + B_t \mathbf{u}_t, \tag{4}$$

$$\Sigma_{t|t-1} = F_t \Sigma_{t-1|t-1} F_t^\top + Q_t. \tag{5}$$

Update Step Innovation (residual):

$$\mathbf{r}_t = \mathbf{z}_t - H_t \hat{\mathbf{x}}_{t|t-1}, \tag{6}$$

$$S_t = H_t \Sigma_{t|t-1} H_t^\top + R_t. \tag{7}$$

Kalman gain:

$$K_t = \Sigma_{t|t-1} H_t^\top S_t^{-1}. \tag{8}$$

Posterior update:

$$\hat{\mathbf{x}}_{t|t} = \hat{\mathbf{x}}_{t|t-1} + K_t \mathbf{r}_t, \tag{9}$$

$$\Sigma_{t|t} = (I - K_t H_t)\Sigma_{t|t-1} \tag{10}$$

We can replace $\Sigma_{t|t} = (I - K_t H_t)\Sigma_{t|t-1}$ with $\Sigma_{t|t} = (I - K_t H_t)\Sigma_{t|t-1}(I - K_t H_t)^\top + K_t R_t K_t^\top$ (Joseph form), as the latter formulation is more robust and ensures that $\Sigma_{t|t}$ remains a positive semi-definite matrix.

The notation $t \mid t-1$ means "at time t given information up to time $t-1$". Thus, $\hat{\mathbf{x}}_{t|t-1}$ is the *predicted* state at time t before observing $\mathbf{z}_t$, while $\hat{\mathbf{x}}_{t|t}$ is the *updated* (filtered) state after incorporating $\mathbf{z}_t$.

Algorithm 2 Kalman Filter for a Linear-Gaussian State-Space Model

Input: $\{F_t, B_t, H_t, Q_t, R_t\}_{t=1}^T$, initial $(\hat{\mathbf{x}}_{0|0}, \Sigma_{0|0})$
for $t = 1$ to T **do**
  Prediction:
  $\hat{\mathbf{x}}_{t|t-1} = F_t \hat{\mathbf{x}}_{t-1|t-1} + B_t \mathbf{u}_t$
  $\Sigma_{t|t-1} = F_t \Sigma_{t-1|t-1} F_t^\top + Q_t$
  Innovation:
  $\mathbf{r}_t = \mathbf{z}_t - H_t \hat{\mathbf{x}}_{t|t-1}$
  $S_t = H_t \Sigma_{t|t-1} H_t^\top + R_t$
  Kalman Gain:
  $K_t = \Sigma_{t|t-1} H_t^\top S_t^{-1}$
  Update:
  $\hat{\mathbf{x}}_{t|t} = \hat{\mathbf{x}}_{t|t-1} + K_t \mathbf{r}_t$
  $\Sigma_{t|t} = (I - K_t H_t)\Sigma_{t|t-1}$ or $\Sigma_{t|t} = (I - K_t H_t)\Sigma_{t|t-1}(I - K_t H_t)^\top + K_t R_t K_t^\top$

Note: In our code implementation, we replace F_t, H_t, Q_t, R_t, B_t with F, H, Q, R, B, assuming they are time-invariant.

Analyze its filtering optimality and numerical stability: compare filtered means/covariances to the Kalman recursion; use Joseph stabilized covariance updates; discuss conditioning number.

Regarding **compare filterd means/convarinces**: According to Wikipedia (kal, 2025), the Kalman filter is described as the best possible linear estimator in the minimum mean-square-error sense. It provides an optimal state estimate when (a) the model perfectly represents the real system, (b) the process noise is white (uncorrelated), and (c) the noise covariances are known exactly. Pei et al. (2019) demonstrated that two key concepts—optimal linear estimation for fusing uncorrelated estimates and best linear unbiased estimation for correlated variables—form the theoretical foundation of Kalman filtering. This perspective

clarifies the assumptions underlying the filter's optimality. For a more detailed derivation of the Kalman filter, see Pei et al. (2019).

Regarding using Joseph stabilized covariance updates: I have added the Joseph covariance update to the original code as well.

Regarding condition number: According to Evangelidis and Parker (2019), the **condition number** of a matrix provides an indication of its numerical stability by measuring how sensitive it is to numerical errors and how close it is to being singular. In their study of Kalman filter implementations, the condition number of the estimation-error covariance matrix P^+ is used as a quantitative measure of robustness against numerical instability. It is defined as

$$\kappa(P^+) = \frac{\sigma_{\max}}{\sigma_{\min}},$$

where $\sigma_{\max}$ and $\sigma_{\min}$ denote the largest and smallest singular values of P^+, respectively. A small condition number (close to one) indicates that the matrix is **well-conditioned**, meaning it is far from singular and that numerical computations will remain stable. Conversely, a large condition number implies an **ill-conditioned** matrix, where small rounding or truncation errors can be amplified, potentially leading to the loss of positive-definiteness in P^+ and, consequently, instability in the Kalman filter. Therefore, monitoring the condition number offers a systematic means to assess the numerical reliability of the covariance matrix and to ensure the Kalman filter operates within stable numerical bounds (Evangelidis and Parker, 2019). For a more detailed description of the study *Quantitative Verification of Numerical Stability for Kalman Filters*, refer to Evangelidis and Parker (2019).

This notebook (`https://github.com/ChunchaoPeter/ml_simulation_research/blob/features_kf_lgssm/kf_lgssm/kalman_filter_analysis_optimality_stability.ipynb`) provides a comprehensive analysis of Kalman filter numerical stability for Linear Gaussian State Space Models (LGSSM). We compare two covariance update formulations: the standard form $\Sigma_{t|t} = (I - K_t H_t)\Sigma_{t|t-1}$ and the Joseph stabilized form $\Sigma_{t|t} = (I - K_t H_t)\Sigma_{t|t-1}(I - K_t H_t)^T + K_t R_t K_t^T$. Using an extended time horizon ($T = 100$) and conditioning number analysis following Evangelidis and Parker (2019),we quantify numerical precision loss via $\log_{10}(\kappa)$. In well-conditioned scenarios, both forms produce equivalent results with $\log_{10}(\kappa) \approx 0.13$ digits of precision loss. However, in ill-conditioned cases (observation noise $R = 10^{-8}$, process noise $Q = 2.0$, ratio $Q/R \approx 2 \times 10^8$), the standard form catastrophically fails—the covariance matrix collapses to zero, yielding undefined conditioning numbers—while the Joseph form maintains stability with $\log_{10}(\kappa) = 0$, preserving positive-definiteness and symmetry. This demonstrates the critical importance of the Joseph formulation for robust Kalman filtering in practical applications with high measurement precision or ill-conditioned dynamics.

Design a nonlinear and non-Gaussian SSM; you can use a stochastic volatility model (example 4 in Doucet and Johansen (2009)) or nonlinear tracking models (e.g., range–bearing observation model).

I have developed the Range–Bearing observation model that is a **state-space model**

with:

- **Linear motion model** (constant velocity in Cartesian coordinates)
- **Non-linear observation model** (polar coordinates: range and bearing)

The non-linearity in the observation function, combined with additive Gaussian noise, produces a **non-Gaussian posterior distribution**, making this model suitable for testing non-linear filtering algorithms such as Extended Kalman Filter (EKF), Unscented Kalman Filter (UKF), and Particle Filter (PF).

State Space: The state at time t is a 4-dimensional vector representing 2D position and velocity:

$$\mathbf{x}_t = \begin{bmatrix} x_t \\ \dot{x}_t \\ y_t \\ \dot{y}_t \end{bmatrix} \in \mathbb{R}^4 \tag{11}$$

where:

- x_t: position in x-direction (horizontal)
- $\dot{x}_t$: velocity in x-direction
- y_t: position in y-direction (vertical)
- $\dot{y}_t$: velocity in y-direction

Motion Model (State Transition): The motion model describes how the state evolves over time using a **linear constant velocity model**:

$$\mathbf{x}_t = A\mathbf{x}_{t-1} + \mathbf{w}_t \tag{12}$$

where $\mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, Q)$ is the **process noise**.

State Transition Matrix:

$$A = \begin{bmatrix} 1 & \Delta t & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \Delta t \\ 0 & 0 & 0 & 1 \end{bmatrix} \in \mathbb{R}^{4\times 4} \tag{13}$$

This matrix implements the discrete-time constant velocity model:

$$x_t = x_{t-1} + \Delta t \cdot \dot{x}_{t-1} + w_{x,t} \tag{14}$$
$$\dot{x}_t = \dot{x}_{t-1} + w_{\dot{x},t} \tag{15}$$
$$y_t = y_{t-1} + \Delta t \cdot \dot{y}_{t-1} + w_{y,t} \tag{16}$$
$$\dot{y}_t = \dot{y}_{t-1} + w_{\dot{y},t} \tag{17}$$

Process Noise Covariance:

$$Q = \begin{bmatrix} \sigma_x^2 & 0 & 0 & 0 \\ 0 & \sigma_{\dot{x}}^2 & 0 & 0 \\ 0 & 0 & \sigma_y^2 & 0 \\ 0 & 0 & 0 & \sigma_{\dot{y}}^2 \end{bmatrix} \in \mathbb{R}^{4\times 4} \tag{18}$$

where:

- σ_x^2, σ_y^2: variance of position noise
- $\sigma_{\dot{x}}^2, \sigma_{\dot{y}}^2$: variance of velocity noise

Observation Model (Measurement): The observation model is **non-linear**, converting Cartesian state to polar coordinates:

$$\mathbf{z}_t = h(\mathbf{x}_t) + \mathbf{v}_t \tag{19}$$

where $\mathbf{v}_t \sim \mathcal{N}(\mathbf{0}, R)$ is the **measurement noise**.

Observation Function:

$$h(\mathbf{x}_t) = \begin{bmatrix} r_t \\ \theta_t \end{bmatrix} = \begin{bmatrix} \sqrt{x_t^2 + y_t^2} \\ \arctan 2(y_t, x_t) \end{bmatrix} \in \mathbb{R}^2 \tag{20}$$

where:

- r_t: **range** (distance from origin to object), $r_t \geq 0$
- θ_t: **bearing** (angle from positive x-axis), $\theta_t \in [-\pi, \pi]$

Range Component: The range (distance) is computed as:

$$r_t = \sqrt{x_t^2 + y_t^2} \tag{21}$$

Bearing Component: The bearing (angle) uses the $\arctan 2$ function for correct quadrant handling:

$$\theta_t = \arctan 2(y_t, x_t) = \begin{cases} \arctan\left(\frac{y_t}{x_t}\right) & \text{if } x_t > 0 \\ \arctan\left(\frac{y_t}{x_t}\right) + \pi & \text{if } x_t < 0, y_t \geq 0 \\ \arctan\left(\frac{y_t}{x_t}\right) - \pi & \text{if } x_t < 0, y_t < 0 \\ +\frac{\pi}{2} & \text{if } x_t = 0, y_t > 0 \\ -\frac{\pi}{2} & \text{if } x_t = 0, y_t < 0 \\ \text{undefined} & \text{if } x_t = 0, y_t = 0 \end{cases} \tag{22}$$

The $\arctan 2$ function returns angles in the range $(-\pi, \pi]$.

Measurement Noise Covariance:

$$R = \begin{bmatrix} \sigma_r^2 & 0 \\ 0 & \sigma_\theta^2 \end{bmatrix} \in \mathbb{R}^{2\times 2} \tag{23}$$

where:

- σ_r^2: variance of range measurement noise
- σ_θ^2: variance of bearing measurement noise (in radians)

Observation Jacobian (for EKF): For the Extended Kalman Filter, we need the **Jacobian matrix** of the observation function:

$$H_t = \left.\frac{\partial h}{\partial \mathbf{x}}\right|_{\mathbf{x}=\mathbf{x}_t} = \begin{bmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial \dot{x}} & \frac{\partial r}{\partial y} & \frac{\partial r}{\partial \dot{y}} \\ \frac{\partial \theta}{\partial x} & \frac{\partial \theta}{\partial \dot{x}} & \frac{\partial \theta}{\partial y} & \frac{\partial \theta}{\partial \dot{y}} \end{bmatrix} \tag{24}$$

Partial Derivatives for Range: For $r = \sqrt{x^2 + y^2}$:

$$\frac{\partial r}{\partial x} = \frac{x}{\sqrt{x^2 + y^2}} = \frac{x}{r} = \cos(\theta) \tag{25}$$

$$\frac{\partial r}{\partial y} = \frac{y}{\sqrt{x^2 + y^2}} = \frac{y}{r} = \sin(\theta) \tag{26}$$

$$\frac{\partial r}{\partial \dot{x}} = 0, \quad \frac{\partial r}{\partial \dot{y}} = 0 \tag{27}$$

Partial Derivatives for Bearing: For $\theta = \arctan 2(y, x)$:

$$\frac{\partial \theta}{\partial x} = \frac{-y}{x^2 + y^2} = \frac{-y}{r^2} = -\frac{\sin(\theta)}{r} \tag{28}$$

$$\frac{\partial \theta}{\partial y} = \frac{x}{x^2 + y^2} = \frac{x}{r^2} = \frac{\cos(\theta)}{r} \tag{29}$$

$$\frac{\partial \theta}{\partial \dot{x}} = 0, \quad \frac{\partial \theta}{\partial \dot{y}} = 0 \tag{30}$$

Complete Jacobian Matrix:

$$H_t = \begin{bmatrix} \cos(\theta_t) & 0 & \sin(\theta_t) & 0 \\ -\frac{\sin(\theta_t)}{r_t} & 0 & \frac{\cos(\theta_t)}{r_t} & 0 \end{bmatrix} \in \mathbb{R}^{2\times 4} \tag{31}$$

The implementation details can be found here:(`https://github.com/ChunchaoPeter/ml_simulation_research/tree/main/ekf_ukf_pf`).

Implement the Extended Kalman Filter (EKF) and Unscented Kalman Filter (UKF). Discuss linearization accuracy limits and sigma-point failures under strong nonlinearity.

First, we consider a nonlinear and non-Gaussian state-space model; the range–bearing observation model above is one example.

State Equation

$$\mathbf{x}_t = f(\mathbf{x}_{t-1}, \mathbf{u}_t) + \mathbf{w}_t \tag{32}$$

where

- $\mathbf{x}_t \in \mathbb{R}^n$ is the state vector at time t,

- $f(\cdot)$ is the (potentially non-linear) state transition function,
- $\mathbf{u}_t$ is the control input (optional),
- $\mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, Q_t)$ is the process noise.

Observation Equation

$$\mathbf{z}_t = h(\mathbf{x}_t) + \mathbf{v}_t \tag{33}$$

where

- $\mathbf{z}_t \in \mathbb{R}^m$ is the observation vector at time t,
- $h(\cdot)$ is the (potentially non-linear) observation function,
- $\mathbf{v}_t \sim \mathcal{N}(\mathbf{0}, R_t)$ is the measurement noise.

Noise Covariances

$$Q_t \in \mathbb{R}^{n\times n} \quad \text{(process noise covariance)}, \tag{34}$$
$$R_t \in \mathbb{R}^{m\times m} \quad \text{(measurement noise covariance)}. \tag{35}$$

Initial Distribution

$$\mathbf{x}_0 \sim \mathcal{N}(\boldsymbol{\mu}_0, \Sigma_0) \tag{36}$$

Extended Kalman Filter (EKF): The Extended Kalman Filter (Pei et al., 2019) handles non-linearity by **linearization** using a first-order Taylor expansion around the current estimate.

Linearization via Jacobians: The EKF linearizes the non-linear functions using Jacobian matrices.

State Transition Jacobian:

$$F_t = \left.\frac{\partial f}{\partial \mathbf{x}}\right|_{\mathbf{x}=\mathbf{x}_{t-1|t-1}} \in \mathbb{R}^{n\times n} \tag{37}$$

This Jacobian is evaluated at the previous filtered state $\mathbf{x}_{t-1|t-1}$.

Observation Jacobian

$$H_t = \left.\frac{\partial h}{\partial \mathbf{x}}\right|_{\mathbf{x}=\mathbf{x}_{t|t-1}} \in \mathbb{R}^{m\times n} \tag{38}$$

This Jacobian is evaluated at the predicted state $\mathbf{x}_{t|t-1}$.

The main difference between the Kalman Filter and the Extended Kalman Filter is that the Extended Kalman Filter uses the Jacobian matrix to linearize nonlinear functions. Everything else is similar, and the pseudocode is shown below.

Algorithm 3 Extended Kalman Filter (EKF)

Input: non-linear functions f, h, noise covariances $\{Q_t, R_t\}_{t=1}^T$, controls $\{\mathbf{u}_t\}_{t=1}^T$, initial $(\hat{\mathbf{x}}_{0|0}, \Sigma_{0|0})$
for $t = 1$ to T **do**
  Prediction:
  $F_t = \left.\frac{\partial f}{\partial \mathbf{x}}\right|_{\mathbf{x}=\hat{\mathbf{x}}_{t-1|t-1}}$
  $\hat{\mathbf{x}}_{t|t-1} = f(\hat{\mathbf{x}}_{t-1|t-1}, \mathbf{u}_t)$
  $\Sigma_{t|t-1} = F_t \Sigma_{t-1|t-1} F_t^\top + Q_t$
  Innovation:
  $H_t = \left.\frac{\partial h}{\partial \mathbf{x}}\right|_{\mathbf{x}=\hat{\mathbf{x}}_{t|t-1}}$
  $\mathbf{r}_t = \mathbf{z}_t - h(\hat{\mathbf{x}}_{t|t-1})$
  $S_t = H_t \Sigma_{t|t-1} H_t^\top + R_t$
  Kalman Gain:
  $K_t = \Sigma_{t|t-1} H_t^\top S_t^{-1}$
  Update:
  $\hat{\mathbf{x}}_{t|t} = \hat{\mathbf{x}}_{t|t-1} + K_t \mathbf{r}_t$
  $\Sigma_{t|t} = (I - K_t H_t)\Sigma_{t|t-1}$ **or** $\Sigma_{t|t} = (I - K_t H_t)\Sigma_{t|t-1}(I - K_t H_t)^\top + K_t R_t K_t^\top$

Unscented Kalman Filter (UKF): The Unscented Kalman Filter (Wan and Van Der Merwe, 2000) handles non-linearity using the **unscented transform**, which propagates carefully chosen **sigma points** through the non-linear functions without requiring Jacobian computation. **Key Idea:** 1. Generate a minimal set of sigma points that capture the mean and covariance of the state distribution; 2. Associate each sigma point with a weight; 3. Transform the sigma points through the non-linear function; 4. Compute a Gaussian approximation from the weighted transformed points.

Sigma Points Generation: For a state with mean $\bar{\mathbf{x}} \in \mathbb{R}^n$ and covariance $\Sigma_{\mathbf{x}} \in \mathbb{R}^{n \times n}$, generate $2n + 1$ sigma points:

$$\boldsymbol{\chi} = \{\boldsymbol{\chi}_i, W_i\}_{i=0}^{2n}, \tag{39}$$

where

$$\boldsymbol{\chi}_0 = \bar{\mathbf{x}}, \tag{40}$$

$$\boldsymbol{\chi}_i = \bar{\mathbf{x}} + \left(\sqrt{(n+\lambda)\Sigma_{\mathbf{x}}}\right)_i, \quad i = 1, 2, \ldots, n, \tag{41}$$

$$\boldsymbol{\chi}_i = \bar{\mathbf{x}} - \left(\sqrt{(n+\lambda)\Sigma_{\mathbf{x}}}\right)_{i-n}, \quad i = n+1, \ldots, 2n, \tag{42}$$

and $\left(\sqrt{(n+\lambda)\Sigma_{\mathbf{x}}}\right)_i$ denotes the i-th row of the matrix square root.

Scaling Parameters:

$$\lambda = \alpha^2(n + \kappa) - n \tag{43}$$

where

- α controls the spread of sigma points around the mean (typically $10^{-3} \leq \alpha \leq 1$),
- κ is a secondary scaling parameter (typically $\kappa = 0$),
- β incorporates prior knowledge of the distribution ($\beta = 2$ is optimal for Gaussian).

Weights for Mean

$$W_0^{(m)} = \frac{\lambda}{n+\lambda}, \tag{44}$$

$$W_i^{(m)} = \frac{1}{2(n+\lambda)}, \quad i = 1, 2, \ldots, 2n. \tag{45}$$

Weights for Covariance

$$W_0^{(c)} = \frac{\lambda}{n+\lambda} + (1 - \alpha^2 + \beta), \tag{46}$$

$$W_i^{(c)} = \frac{1}{2(n+\lambda)}, \quad i = 1, 2, \ldots, 2n. \tag{47}$$

Unscented Transform: Given sigma points $\{\boldsymbol{\chi}_i\}_{i=0}^{2n}$ and a non-linear function $g(\cdot)$:

1. **Transform sigma points:**

$$\boldsymbol{\gamma}_i = g(\boldsymbol{\chi}_i), \quad i = 0, 1, \ldots, 2n. \tag{48}$$

2. **Compute mean:**

$$\bar{\mathbf{y}} = \sum_{i=0}^{2n} W_i^{(m)} \boldsymbol{\gamma}_i. \tag{49}$$

3. **Compute covariance:**

$$\Sigma_{\mathbf{y}} = \sum_{i=0}^{2n} W_i^{(c)} (\boldsymbol{\gamma}_i - \bar{\mathbf{y}})(\boldsymbol{\gamma}_i - \bar{\mathbf{y}})^\top + \text{noise covariance}. \tag{50}$$

The implementation details can be found here:(`https://github.com/ChunchaoPeter/ml_simulation_research/tree/main/ekf_ukf_pf`).

Regarding Discuss linearization accuracy limits and sigma-point failures under strong nonlinearity:

The EKF suffers linearization accuracy limits because it linearizes the nonlinear system using Jacobians, giving only first-order accurate propagation of mean/covariance. This introduces large errors when curvature is strong and may cause divergence (Wan and Van Der Merwe, 2000). The UKF avoids Jacobians by using sigma points that capture mean

Algorithm 4 Unscented Kalman Filter (UKF)

Input: non-linear functions f, h, noise covariances $\{Q_t, R_t\}_{t=1}^T$, controls $\{\mathbf{u}_t\}_{t=1}^T$, initial $(\hat{\mathbf{x}}_{0|0}, \Sigma_{0|0})$, UKF parameters α, β, κ
$\lambda = \alpha^2(n+\kappa) - n$
$W_0^{(m)} = \lambda/(n+\lambda)$
$W_0^{(c)} = \lambda/(n+\lambda) + (1-\alpha^2+\beta)$
$W_i^{(m)} = W_i^{(c)} = 1/(2(n+\lambda))$ for $i = 1, \ldots, 2n$
for $t = 1$ to T **do**
    Prediction:
    $\boldsymbol{\chi}_{0,t-1} = \hat{\mathbf{x}}_{t-1|t-1}$
    $\boldsymbol{\chi}_{i,t-1} = \hat{\mathbf{x}}_{t-1|t-1} + \left(\sqrt{(n+\lambda)\Sigma_{t-1|t-1}}\right)_i$, $i = 1, \ldots, n$
    $\boldsymbol{\chi}_{i,t-1} = \hat{\mathbf{x}}_{t-1|t-1} - \left(\sqrt{(n+\lambda)\Sigma_{t-1|t-1}}\right)_{i-n}$, $i = n+1, \ldots, 2n$
    $\boldsymbol{\chi}_{i,t|t-1} = f(\boldsymbol{\chi}_{i,t-1}, \mathbf{u}_t)$, $i = 0, \ldots, 2n$
    $\hat{\mathbf{x}}_{t|t-1} = \sum_{i=0}^{2n} W_i^{(m)} \boldsymbol{\chi}_{i,t|t-1}$
    $\Sigma_{t|t-1} = \sum_{i=0}^{2n} W_i^{(c)} (\boldsymbol{\chi}_{i,t|t-1} - \hat{\mathbf{x}}_{t|t-1})(\boldsymbol{\chi}_{i,t|t-1} - \hat{\mathbf{x}}_{t|t-1})^\top + Q_t$
    Update:
    $\mathbf{z}_{i,t|t-1} = h(\boldsymbol{\chi}_{i,t|t-1})$, $i = 0, \ldots, 2n$
    $\hat{\mathbf{z}}_{t|t-1} = \sum_{i=0}^{2n} W_i^{(m)} \mathbf{z}_{i,t|t-1}$
    $S_t = \sum_{i=0}^{2n} W_i^{(c)} (\mathbf{z}_{i,t|t-1} - \hat{\mathbf{z}}_{t|t-1})(\mathbf{z}_{i,t|t-1} - \hat{\mathbf{z}}_{t|t-1})^\top + R_t$
    $\Sigma_{\mathbf{xz}}^{t|t-1} = \sum_{i=0}^{2n} W_i^{(c)} (\boldsymbol{\chi}_{i,t|t-1} - \hat{\mathbf{x}}_{t|t-1})(\mathbf{z}_{i,t|t-1} - \hat{\mathbf{z}}_{t|t-1})^\top$
    $K_t = \Sigma_{\mathbf{xz}}^{t|t-1} S_t^{-1}$
    $\hat{\mathbf{x}}_{t|t} = \hat{\mathbf{x}}_{t|t-1} + K_t(\mathbf{z}_t - \hat{\mathbf{z}}_{t|t-1})$
    $\Sigma_{t|t} = \Sigma_{t|t-1} - K_t S_t K_t^\top$

and covariance exactly and propagate them through the true nonlinear dynamics, giving third-order accuracy (Wan and Van Der Merwe, 2000).

Under strong nonlinearity, sigma-point methods can break down because the nonlinear transformation introduces large higher-order terms that the sigma-point approximation cannot capture. In the standard UKF, this issue is aggravated in higher-dimensional systems (dimension greater than three), where the central sigma point receives a large negative weight, making the covariance update numerically unstable. When strong nonlinear functions amplify these unmodelled higher-order components, the sigma-point approximation can yield distorted mean and covariance estimates, leading to significant errors or divergence (Chang et al., 2013).

Implement a standard particle filter for your model (do not use `tfp.experimental.mcmc.particle_filter`). Visualize and discuss issues such as particle degeneracy.

Particle Filter: The Particle Filter (Doucet and Johansen, 2009) is a Sequential Monte Carlo (SMC) method that handles non-linearity and non-Gaussianity by representing the posterior distribution using a set of weighted particles. Unlike EKF and UKF which

approximate the posterior as a Gaussian, the particle filter can represent arbitrary distributions. **Key Idea:** Represent the filtering distribution $p(\mathbf{x}_t|\mathbf{z}_{1:t})$ using N weighted particles:

$$p(\mathbf{x}_t|\mathbf{z}_{1:t}) \approx \sum_{i=1}^{N} w_t^i \delta(\mathbf{x}_t - \mathbf{X}_t^i) \tag{51}$$

where $\{\mathbf{X}_t^i, w_t^i\}_{i=1}^N$ are particles and their associated weights.

Hidden Markov Model formulation: Here I begin indexing at 0 to maintain consistency with the KF, EKF, and UKF. Alternatively, the formulation may start at index 1 Doucet and Johansen (2009). For a state sequence $\mathbf{X}_{0:t} = \{\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_t\}$ and observations $\mathbf{Z}_{1:t} = \{\mathbf{z}_1, \ldots, \mathbf{z}_t\}$:

Prior (Markov property):

$$p(\mathbf{x}_{0:t}) = \mu(\mathbf{x}_0) \prod_{k=1}^{t} f(\mathbf{x}_k|\mathbf{x}_{k-1}) \tag{52}$$

Likelihood (Conditional independence):

$$p(\mathbf{z}_{1:t}|\mathbf{x}_{0:t}) = \prod_{k=1}^{t} g(\mathbf{z}_k|\mathbf{x}_k) \tag{53}$$

Posterior (Bayes' rule):

$$p(\mathbf{x}_{0:t}|\mathbf{z}_{1:t}) = \frac{p(\mathbf{x}_{0:t}, \mathbf{z}_{1:t})}{p(\mathbf{z}_{1:t})} = \frac{p(\mathbf{x}_{0:t})p(\mathbf{z}_{1:t}|\mathbf{x}_{0:t})}{p(\mathbf{z}_{1:t})} \tag{54}$$

where

$$p(\mathbf{x}_{0:t}, \mathbf{z}_{1:t}) = p(\mathbf{x}_{0:t})p(\mathbf{z}_{1:t}|\mathbf{x}_{0:t}), \tag{55}$$

$$p(\mathbf{z}_{1:t}) = \int p(\mathbf{x}_{0:t}, \mathbf{z}_{1:t}) d\mathbf{x}_{0:t}. \tag{56}$$

Filtering Distribution: The goal is to compute the marginal filtering distribution:

$$p(\mathbf{x}_t|\mathbf{z}_{1:t}) = \int p(\mathbf{x}_{0:t}|\mathbf{z}_{1:t}) d\mathbf{x}_{0:t-1} \tag{57}$$

The filtering distribution can be computed recursively:

Prediction Step:

$$p(\mathbf{x}_t|\mathbf{z}_{1:t-1}) = \int f(\mathbf{x}_t|\mathbf{x}_{t-1}) p(\mathbf{x}_{t-1}|\mathbf{z}_{1:t-1}) d\mathbf{x}_{t-1} \tag{58}$$

Update Step:

$$p(\mathbf{x}_t|\mathbf{z}_{1:t}) = \frac{g(\mathbf{z}_t|\mathbf{x}_t)p(\mathbf{x}_t|\mathbf{z}_{1:t-1})}{p(\mathbf{z}_t|\mathbf{z}_{1:t-1})} \tag{59}$$

For non-linear or non-Gaussian systems, these integrals have no analytic solution. The particle filter approximates these distributions using Monte Carlo sampling.

Based on Sequential Importance Sampling (Doucet and Johansen, 2009), we use a sequential proposal that factorizes as:

$$q(\mathbf{x}_{0:t}|\mathbf{z}_{1:t}) = q(\mathbf{x}_0) \prod_{k=1}^{t} q(\mathbf{x}_k|\mathbf{x}_{k-1}, \mathbf{z}_k) \tag{60}$$

This allows us to sample new particles $\mathbf{X}_t^i \sim q(\mathbf{x}_t|\mathbf{X}_{t-1}^i, \mathbf{z}_t)$ and update weights recursively:

$$w_t^i \propto w_{t-1}^i \frac{g(\mathbf{z}_t|\mathbf{X}_t^i) f(\mathbf{X}_t^i|\mathbf{X}_{t-1}^i)}{q(\mathbf{X}_t^i|\mathbf{X}_{t-1}^i, \mathbf{z}_t)} \tag{61}$$

In Eq. (61), w_t^i denotes the *unnormalised importance weight* of particle i at time t. It is updated recursively to correct for the discrepancy between the proposal distribution and the true filtering model. A larger weight indicates that the particle's predicted state X_t^i is more consistent with the new observation z_t.

Prior Proposal: A common choice is the **prior proposal**:

$$q(\mathbf{x}_0) = \mu(\mathbf{x}_0), \tag{62}$$

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{z}_t) = f(\mathbf{x}_t|\mathbf{x}_{t-1}) \tag{63}$$

With this choice, the weight update simplifies to:

$$w_t^i \propto w_{t-1}^i \frac{g(\mathbf{z}_t|\mathbf{X}_t^i) f(\mathbf{X}_t^i|\mathbf{X}_{t-1}^i)}{f(\mathbf{X}_t^i|\mathbf{X}_{t-1}^i)} = w_{t-1}^i \cdot g(\mathbf{z}_t|\mathbf{X}_t^i) \tag{64}$$

Resampling: SIS suffers from exponentially increasing variance as t grows, due to increasingly unbalanced importance weights. Resampling in SMC methods helps mitigate—but not eliminate—this degeneracy. One common resampling method is *multinomial resampling*. We draw an ancestry vector $\mathbf{A}^{1:N}$, where each A^i is sampled independently as

$$A^i \sim \text{Categorical}(w_t^1, w_t^2, \ldots, w_t^N), \qquad i = 1, \ldots, N. \tag{65}$$

Then set:

$$\tilde{\mathbf{X}}_t^i = \mathbf{X}_t^{A^i} \quad \text{(resampled particles)}, \tag{66}$$

$$\tilde{w}_t^i = \frac{1}{N} \quad \text{(uniform weights after resampling)}. \tag{67}$$

The implementation details can be found here: (`https://github.com/ChunchaoPeter/ml_simulation_research/tree/main/ekf_ukf_pf`).

Regarding visualize and discuss issues such as particle degeneracy:

Figure 1 illustrates the path degeneracy phenomenon in particle filters. Although the particle population may appear diverse at the final time step, their ancestral lineages collapse

Algorithm 5 Particle Filter with Prior Proposal and Resampling

Input: non-linear functions f, h, noise samplers for Q_t, R_t, initial distribution $\mu(\mathbf{x}_0)$, number of particles N
Initialization (t = 0):
for $i = 1$ to N **do**
    Sample $\mathbf{X}_0^i \sim \mu(\mathbf{x}_0)$
    Set $w_0^i = 1/N$
$\hat{\mathbf{x}}_0 = \sum_{i=1}^N w_0^i \mathbf{X}_0^i$

for $t = 1$ to T **do**
    Prediction:
    for $i = 1$ to N **do**
        Sample $\mathbf{X}_t^i \sim f(\mathbf{x}_t | \mathbf{X}_{t-1}^i, \mathbf{u}_t)$ ▷ Propagate particles through dynamics

    Update:
    for $i = 1$ to N **do**
        Compute $\tilde{w}_t^i = w_{t-1}^i \cdot g(\mathbf{z}_t | \mathbf{X}_t^i)$ ▷ Weight by observation likelihood
    Normalize: $w_t^i = \tilde{w}_t^i / \sum_{j=1}^N \tilde{w}_t^j$

    State Estimation:
    $\hat{\mathbf{x}}_t = \sum_{i=1}^N w_t^i \mathbf{X}_t^i$

    Resampling:
    for $i = 1$ to N **do**
        Draw $A^i \sim \text{Categorical}(w_t^1, \ldots, w_t^N)$
        Set $\mathbf{X}_t^i = \mathbf{X}_t^{A^i}$ ▷ Resample particles
    Set $w_t^i = 1/N$ for all i ▷ Reset to uniform weights

very quickly after just a few rounds of resampling. This means that, going backward in time, nearly all particles descend from a single early ancestor. One of the reason is that Resampling repeatedly duplicates particles with high weights while discarding those with low weights. Over many iterations, this causes most particles to descend from only a few early ancestors, leading to a rapid loss of genealogical diversity.

Compare PF and EKF/UKF performance. How to evaluate your SSMs? What metrics can you use? In practice we care about runtime and memory; compare runtime and peak memory (CPU/GPU RAM) for each SSM.

State Space Model Comparison and Evaluation: We evaluated three state estimation algorithms—Extended Kalman Filter (EKF), Unscented Kalman Filter (UKF), and Particle Filter (PF)—on a range-bearing tracking problem with mildly non-linear dynamics. State space models are evaluated using three categories of metrics: accuracy metrics (position RMSE, velocity RMSE, and average position error over time), computational performance metrics (runtime and peak memory usage).

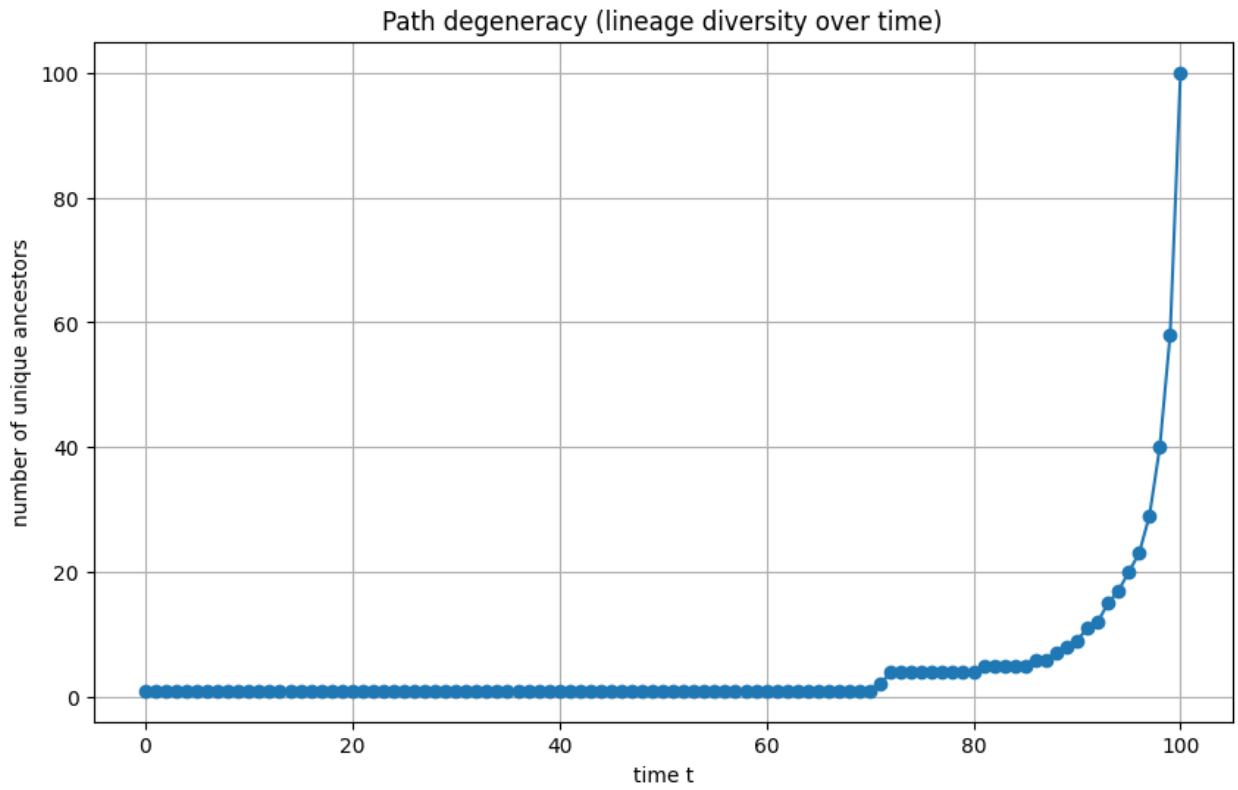

Figure 1: Particle Degeneracy

Accuracy Performance: In terms of accuracy, EKF and UKF achieved nearly identical performance with position RMSE of 2.33 and 2.34 respectively, and velocity RMSE of approximately 0.42 for both filters. The Particle Filter with 100 particles demonstrated inferior accuracy with position RMSE of 4.54 and velocity RMSE of 0.53. For this mildly non-linear system, the linearization-based EKF and the sigma-point-based UKF both provide accurate state estimates, while PF accuracy is limited by the relatively small particle count used in the comparison.

Runtime Performance: The computational runtime differs dramatically across the three methods. UKF was the fastest with a runtime of 3.18 seconds, followed closely by EKF at 3.46 seconds. In stark contrast, the Particle Filter required 102.69 seconds to process the same trajectory. This represents a speedup factor of approximately 32× for UKF and 30× for EKF compared to PF. The significant computational advantage of EKF and UKF stems from their parametric representation of the posterior distribution as a Gaussian, requiring only mean and covariance updates, whereas PF must propagate and resample hundreds of particles at each time step.

Memory Efficiency: Peak memory usage shows similar disparities. UKF was the most memory-efficient algorithm, using only 1.97 MB of RAM, which represents just 23.9% of the Particle Filter's memory footprint. EKF required 3.45 MB (41.8% of PF memory), while PF consumed 8.25 MB. The memory requirements of PF scale linearly with the number of particles, making it particularly demanding for real-time applications or resource-constrained environments. UKF's superior memory efficiency compared to EKF is noteworthy, as it

achieves this while maintaining derivative-free operation through sigma point sampling.

Note: All experiments for this question were performed on a 2017 15-inch MacBook Pro equipped with a 3.1 GHz quad-core Intel Core i7 processor, Radeon Pro 560 graphics, 16 GB of LPDDR3 memory, and running macOS Ventura 13.7.8. For more detail, see the notebook at `https://github.com/ChunchaoPeter/ml_simulation_research/blob/main/ekf_ukf_pf/compare_pf_ekf_ukf.ipynb`

Study the Exact Daum–Huang (EDH) flow and Local Exact Daum–Huang (LEDH) flow (see Daum et al. (2010); Daum and Huang (2011)), and the invertible particle flow particle filter (PF-PF) framework (see Li and Coates (2017)). Replicate the main results in Li and Coates (2017).

Daum et al. (2010); Daum and Huang (2011) introduce particle flow as an alternative to discrete Bayesian weight updates, formulating the measurement update as a continuous transformation of the probability density governed by a partial differential equation. In theory, particle flow methods avoid particle degeneracy, remove the need for resampling, and scale more effectively to high-dimensional state spaces. In this section, we describe a special case of particle flow underlying the Exact Daum–Huang (EDH), Local Exact Daum–Huang (LEDH), and the invertible particle-flow particle filter (PF-PF) framework.

The system is defined as

$$\mathbf{x}_k = f_k(\mathbf{x}_{k-1}) + \mathbf{w}_k, \tag{68}$$

$$\mathbf{y}_k = \gamma_k(\mathbf{x}_k) + \mathbf{v}_k, \tag{69}$$

where

- $\mathbf{x}_k \in \mathbb{R}^{d_x \times 1}$ is the hidden state,
- $\mathbf{y}_k \in \mathbb{R}^{d_y \times 1}$ is the observation,
- $\mathbf{w}_k$ and $\mathbf{v}_k$ denote process and observation noise.

Here, f_k is assumed to be linear such that

$$\mathbf{x}_k = \mathbf{F}_k \mathbf{x}_{k-1} + \mathbf{w}_k. \tag{70}$$

The goal is to recursively estimate the system state $\mathbf{x}_k$ at each time step k. Based on Bayesian theory, the unnormalized posterior is given by

$$\tilde{p}(\mathbf{x}_k \mid \mathbf{y}_{1:k}) = p(\mathbf{y}_k \mid \mathbf{x}_k)\, p(\mathbf{x}_k \mid \mathbf{y}_{1:k-1}). \tag{71}$$

We define

$$g(\mathbf{x}_k) = p(\mathbf{x}_k \mid \mathbf{y}_{1:k-1}), \tag{72}$$

$$h(\mathbf{x}_k) = p(\mathbf{y}_k \mid \mathbf{x}_k), \tag{73}$$

as the predictive prior and the likelihood, respectively.

Following Daum et al. (2010), we introduce a homotopy function $\phi(\mathbf{x}_k, \lambda)$:

$$\phi(\mathbf{x}_k, \lambda) = \log g(\mathbf{x}_k) + \lambda \log h(\mathbf{x}_k), \qquad \lambda \in [0, 1]. \tag{74}$$

Two special cases are of particular interest:

$$\lambda = 0 \Rightarrow \phi(\mathbf{x}_k, 0) = \log g(\mathbf{x}_k), \tag{75}$$

$$\lambda = 1 \Rightarrow \phi(\mathbf{x}_k, 1) = \log \tilde{p}(\mathbf{x}_k \mid \mathbf{y}_{1:k}). \tag{76}$$

Intuitively, λ acts as a continuation parameter that smoothly deforms the prior density into the posterior density. At $\lambda = 0$, the density corresponds to the prior, while at $\lambda = 1$, it coincides with the posterior. This continuous deformation naturally motivates the introduction of a *particle flow*, which transports particles from the prior to the posterior via a continuous transformation.

For the Exact Daum–Huang (EDH) flow, suppose that the particle dynamics are governed by the ordinary differential equation

$$\frac{d\mathbf{x}}{d\lambda} = \psi(\mathbf{x}, \lambda). \tag{77}$$

By invoking the Fokker–Planck equation, the flow field $\psi(\mathbf{x}, \lambda)$ is required to satisfy the following relationship (Ding and Coates, 2012):

$$\frac{\partial \phi}{\partial \mathbf{x}} \psi(\mathbf{x}, \lambda) + \log(h) = -\operatorname{Tr}\left(\frac{\partial \psi}{\partial \mathbf{x}}\right). \tag{78}$$

A solution to this equation defines the *exact* particle flow that transports the probability density continuously from the prior to the posterior.

When both $\log g$ and $\log h$ can be expressed as polynomials in the state variables—such as in Gaussian and other exponential-family distributions—the differential equation $\frac{d\mathbf{x}}{d\lambda}$ admits a closed-form solution derived from 78. In the linear Gaussian setting, let $\bar{\mathbf{x}}$ represent the predicted value of $\mathbf{x}$ and let P denote the associated prediction error covariance. The measurement process is assumed to follow

$$y_k = H\mathbf{x_k} + \mathbf{v}_k,$$

where H is the measurement matrix and R is the covariance of the measurement noise. Under these assumptions, Daum and Huang obtain the following affine flow expression Daum et al. (2010); Ding and Coates (2012):

$$\frac{d\mathbf{x}}{d\lambda} = A(\lambda)\mathbf{x} + b(\lambda). \tag{79}$$

The matrix-valued coefficient $A(\lambda)$ is given by

$$A(\lambda) = -\frac{1}{2} P H^\top \big(\lambda H P H^\top + R\big)^{-1} H, \tag{80}$$

and the corresponding offset term $b(\lambda)$ is defined as

$$b(\lambda) = \big(I + 2\lambda A(\lambda)\big) \left[\big(I + \lambda A(\lambda)\big) P H^\top R^{-1} z + A(\lambda)\bar{\mathbf{x}}\right]. \tag{81}$$

In the case of nonlinear measurement models, local approximations can be introduced via Taylor expansion. The predicted mean $\bar{\mathbf{x}}$ may be estimated directly from the particle ensemble, while linearizing the measurement function yields an approximate measurement matrix $H_{\bar{\mathbf{x}}}$.

Algorithm 6 Original Exact Flow Daum–Huang Filter (EDH)

Initialization: Draw $\{\mathbf{x}_0^i\}_{i=1}^N$ from the prior $p(\mathbf{x}_0)$
Set $\hat{\mathbf{x}}_0$ and $\mathbf{m}_0$ as the mean; P_0 as the covariance matrix
for $k = 1$ to T **do**
    Propagate particles:

$$\mathbf{x}_{k|k-1}^i = f_k(\mathbf{x}_{k-1}^i) + \mathbf{v}_k$$

    Calculate the mean value $\bar{\mathbf{x}}_k$
    Apply UKF/EKF prediction:

$$(\mathbf{m}_{k-1|k-1}, P_{k-1|k-1}) \to (\mathbf{m}_{k|k-1}, P_{k|k-1})$$

    for $j = 1, \ldots, N_\lambda$ **do**
        Set $\lambda = j\Delta\lambda$
        Calculate $H_{\mathbf{x}}$ by linearizing $\gamma_k(\cdot)$ at $\bar{\mathbf{x}}_k$
        Calculate A and b from 80 and 81 using $P_{k|k-1}$, $\bar{\mathbf{x}}_k$, and $H_{\mathbf{x}}$
        for $i = 1, \ldots, N$ **do**
            Evaluate $\dfrac{d\mathbf{x}_k^i}{d\lambda}$ for each particle from 79
            Migrate particles:

$$\mathbf{x}_k^i = \mathbf{x}_k^i + \Delta\lambda \cdot \frac{d\mathbf{x}_k^i}{d\lambda}$$

        Re-evaluate $\bar{\mathbf{x}}_k$ using the updated particles $\{\mathbf{x}_k^i\}$
    Apply UKF/EKF update:

$$(\mathbf{m}_{k|k-1}, P_{k|k-1}) \to (\mathbf{m}_{k|k}, P_{k|k})$$

    Estimate $\hat{\mathbf{x}}_k$ from the particles $\{\mathbf{x}_k^i\}$ using $P_{k|k}$
    Optional: redraw particles $\mathbf{x}_k^i \sim \mathcal{N}(\hat{\mathbf{x}}_k, P_{k|k})$

The local EDH (Ding and Coates, 2012) closely resembles the standard EDH formulation. As implied by the term local, it introduces a key modification whereby the measurement function is linearized for each particle, yielding $H_{\mathbf{x}_i}$, A^i, and b^i. These modifications are summarized in Algorithm 7.

Algorithm 7 Modified Local Exact Flow Daum–Huang Filter (LEDH) (replaces lines 7–17 of Algorithm 6)

1: **for** $j = 1, \ldots, N_\lambda$ **do**
2: Set $\lambda = j\Delta\lambda$;
3: **for** $i = 1, \ldots, N$ **do**
4: Calculate $H_{\mathbf{x}_i}$ by linearizing $\gamma_k(\cdot)$ at $\mathbf{x}^i$;
5: Calculate A^i and b^i from 80 and 81 using $P_{k|k-1}$, $\bar{\mathbf{x}}$ and $H_{\mathbf{x}_i}$;
6: Evaluate $\dfrac{d\mathbf{x}_k^i}{d\lambda}$ for each particle from 79;
7: Migrate particles:

$$\mathbf{x}_k^i = \mathbf{x}_k^i + \Delta\lambda \cdot \frac{d\mathbf{x}_k^i}{d\lambda};$$

8: Re-evaluate $\bar{\mathbf{x}}_k$ using updated particles $\mathbf{x}_k^i$.
9: Apply UKF/EKF update:

$$(\mathbf{m}_{k|k-1}, P_{k|k-1}) \to (\mathbf{m}_{k|k}, P_{k|k});$$

10: Estimate $\hat{\mathbf{x}}_k$ from the particles $\mathbf{x}_k^i$ using $P_{k|k}$;
11: Set $\mathbf{m}_{k|k} = \hat{\mathbf{x}}_k$;
12: **Optional:** redraw particles $\mathbf{x}_k^i \sim \mathcal{N}(\hat{\mathbf{x}}_k, P_{k|k})$;

Regarding the invertible particle flow particle filter (PF-PF) framework (Li and Coates, 2017), deterministic particle flows are embedded within a standard particle filtering scheme by treating the particle flow as an invertible deterministic proposal. Specifically, an invertible mapping

$$\mathbf{x}_k^i = T(\mathbf{x}_{k-1}^i, \mathbf{z}_k)$$

is constructed to transport particles from the predictive prior to the posterior distribution. This mapping induces a proposal density of the form

$$q(\mathbf{x}_k^i \mid \mathbf{x}_{k-1}^i, \mathbf{z}_k) = \frac{p(\mathbf{x}_{k-1}^i \mid \mathbf{z}_{1:k-1})}{\left|\det \nabla_{\mathbf{x}} T(\mathbf{x}_{k-1}^i, \mathbf{z}_k)\right|},$$

which can be evaluated exactly through the associated Jacobian determinant. Within this framework, particle filtering and particle flow are naturally combined, allowing the particle flow to serve as an efficient proposal mechanism. This idea is further applied to the EDH and LEDH particle flow formulations in order to assess their performance within the PF-PF framework.

Algorithm 8 Particle Flow Particle Filtering (EDH)

Initialization: Draw $\{\mathbf{x}_0^i\}_{i=1}^N$ from the prior $p_0(\mathbf{x})$. Set $\hat{\mathbf{x}}_0$ and $\hat{P}_0$ to be the mean and covariance of $p_0(\mathbf{x})$.
Set $\{w_0^i\}_{i=1}^N = \frac{1}{N}$.
for $k = 1$ to T **do**
    Apply EKF/UKF prediction to estimate P

$$(\hat{\mathbf{x}}_{k-1}, P_{k-1}) \to (\mathbf{m}_{k|k-1}, P)$$

    for $i = 1, \ldots, N$ **do**
        Propagate particles: $\boldsymbol{\eta}_0^i = f_k(\mathbf{x}_{k-1}) + \mathbf{w}_k$
        Set $\boldsymbol{\eta}_1^i = \boldsymbol{\eta}_0^i$
    Compute flow linearization point: $\bar{\boldsymbol{\eta}}_0 = f_k(\hat{\mathbf{x}}_{k-1}), \quad \bar{\boldsymbol{\eta}} = \bar{\boldsymbol{\eta}}_0$
    Set $\lambda = 0$
    for $j = 1, \ldots, N_\lambda$ **do**
        $\lambda \leftarrow \lambda + \epsilon_j$
        Compute shared flow parameters $A^j(\lambda)$ and $b^j(\lambda)$ from 80 and 81, linearized at $\bar{\boldsymbol{\eta}}$
        Migrate mean: $\bar{\boldsymbol{\eta}} \leftarrow \bar{\boldsymbol{\eta}} + \epsilon_j \big(A^j(\lambda)\bar{\boldsymbol{\eta}} + b^j(\lambda)\big)$
        for $i = 1, \ldots, N$ **do**
            Migrate particles: $\boldsymbol{\eta}_1^i \leftarrow \boldsymbol{\eta}_1^i + \epsilon_j \big(A^j(\lambda)\boldsymbol{\eta}_1^i + b^j(\lambda)\big)$
    for $i = 1, \ldots, N$ **do**
        Set $\mathbf{x}_k^i = \boldsymbol{\eta}_1^i$
        Update weights: $w_k^i = \frac{p(\mathbf{x}_k^i|\mathbf{x}_{k-1}^i)\, p(\mathbf{z}_k|\mathbf{x}_k^i)}{p(\boldsymbol{\eta}_0^i|\mathbf{x}_{k-1}^i)} w_{k-1}^i$
    Normalize $\{w_k^i\}_{i=1}^N$
    Measurement update: $(\mathbf{m}_{k|k-1}, P_{k|k-1}) \to (\mathbf{m}_{k|k}, P_k)$
    Estimate: $\hat{\mathbf{x}}_k = \sum_{i=1}^N w_k^i \mathbf{x}_k^i$
    Optional: Resample $\{\mathbf{x}_k^i, w_k^i\}_{i=1}^N$

For simplicity, in Algorithms 8 and 9, we focus on illustrating the core idea of the Particle Flow Particle Filter (PF-PF) with EDH and LEDH. Accordingly, the matrices A and b are computed using Equations 80–81. A more detailed formulation of these functions is provided in the original paper (Li and Coates, 2017). In addition, the original work presents a rigorous theoretical justification for the use of invertible particle flows within the PF-PF framework.

The code for each algorithm is implemented in the `deterministic_kernel_flows` folder of our GitHub repository. The file `edh.py` implements both the EDH and LEDH algorithms, with usage demonstrated in the notebook `edh_ledh_demo.ipynb`. The files `pfpf_edh.py` and `pfpf_ledh.py` implement the Particle Flow Particle Filter (PF-PF) with EDH and LEDH, respectively. Demonstrations of these algorithms are provided in `pfpf_edh_demo.ipynb` and `pfpf_ledh_demo_with_comparison.ipynb`. To replicate the main results in Li and Coates (2017), we additionally provide `acoustic_function.py`, which contains the acoustic tracking

Algorithm 9 Particle Flow Particle Filtering (LEDH)

Initialization: Draw $\{\mathbf{x}_0^i\}_{i=1}^N$ from the prior $p_0(\mathbf{x})$. Set $\hat{\mathbf{x}}_0$ and $\hat{P}_0$ to be the mean and covariance of $p_0(\mathbf{x})$.
Set $\{w_0^i\}_{i=1}^N = \frac{1}{N}$.
for $k = 1$ to T **do**
  for $i = 1, \ldots, N$ **do**
    Apply EKF/UKF prediction: $(\mathbf{x}_{k-1}^i, P_{k-1}^i) \to \mathbf{m}_{k|k-1}^i, P^i)$
    Initialize flow: $\bar{\boldsymbol{\eta}}^i = f_k(\mathbf{x}_{k-1}^i), \quad \boldsymbol{\eta}_0^i = f_k(\mathbf{x}_{k-1}^i) + \mathbf{w}_k, \quad \boldsymbol{\eta}_1^i = \boldsymbol{\eta}_0^i, \quad \theta^i = 1$
    Set $\bar{\boldsymbol{\eta}}_0^i = f_k(\mathbf{x}_{k-1}^i)$
  Set $\lambda = 0$
  for $j = 1, \ldots, N_\lambda$ **do**
    $\lambda \leftarrow \lambda + \epsilon_j$
    for $i = 1, \ldots, N$ **do**
      Compute particle-wise flow parameters $A_j^i(\lambda)$ and $b_j^i(\lambda)$ from 80 - 81 linearized at $\bar{\boldsymbol{\eta}}^i$
      Migrate mean: $\bar{\boldsymbol{\eta}}^i \leftarrow \bar{\boldsymbol{\eta}}^i + \epsilon_j \big(A_j^i(\lambda)\bar{\boldsymbol{\eta}}^i + b_j^i(\lambda)\big)$
      Migrate particles: $\boldsymbol{\eta}_1^i \leftarrow \boldsymbol{\eta}_1^i + \epsilon_j \big(A_j^i(\lambda)\boldsymbol{\eta}_1^i + b_j^i(\lambda)\big)$
      Update Jacobian determinant: $\theta^i \leftarrow \theta^i \left|\det\big(I + \epsilon_j A_j^i(\lambda)\big)\right|$
  for $i = 1, \ldots, N$ **do**
    Set $\mathbf{x}_k^i = \boldsymbol{\eta}_1^i$
    Update weights: $w_k^i = \frac{p(\mathbf{x}_k^i|\mathbf{x}_{k-1}^i)\, p(y_k|\mathbf{x}_k^i)\, \theta^i}{p(\boldsymbol{\eta}_0^i|\mathbf{x}_{k-1}^i)} w_{k-1}^i$
  Normalize $\{w_k^i\}_{i=1}^N$
  for $i = 1, \ldots, N$ **do**
    Apply EKF/UKF update: $(\mathbf{m}_{k|k-1}^i, P^i) \to (\mathbf{m}_{k|k}^i, P_k^i)$
  Estimate: $\hat{\mathbf{x}}_k = \sum_{i=1}^N w_k^i \mathbf{x}_k^i$
  Optional: Resample $\{\mathbf{x}_k^i, P_k^i, w_k^i\}_{i=1}^N$

model, along with a corresponding demonstration notebook `acoustic_function_demo.ipynb`. Further implementation details can be found in the relevant code within the GitHub repository.

Implement the kernel-embedded particle flow filter in an RKHS following Hu and Van Leeuwen (2021). Then compare the scalar kernel and diagonal matrix-valued kernel. Use experiments to demonstrate that matrix-valued kernels can prevent collapse of observed-variable marginals in high dimensions; plot similar figures as in Figures 2–3 of Hu and Van Leeuwen (2021).

Particle Flow Filtering (PFF) (Hu and Van Leeuwen, 2021) combines ideas from particle filtering and optimal transport. The particle evolution is defined through a continuous flow governed by the ordinary differential equation

$$\frac{d\mathbf{x}_{k,s}}{ds} = f_{k,s}(\mathbf{x}_{k,s}), \qquad s \in [0, \infty), \quad \mathbf{x}_{k,s} \in \mathbb{R}^{n_x}, \tag{82}$$

where $\mathbf{x}_{k,s}$ denotes the state of a particle at discrete time index k and pseudo-time s. The pseudo-time variable s plays a role analogous to the homotopy (or λ) parameter commonly used in particle flow formulations, such as those appearing in Algorithm 9.

Let

$$q_{k,0}(\mathbf{x}) = p(\mathbf{x}) \quad \text{(prior)}, \qquad q_{k,\infty}(\mathbf{x}) = p(\mathbf{x} \mid \mathbf{y}) \quad \text{(posterior)}. \tag{83}$$

The goal is to choose a suitable flow field $f_{k,s}$ such that the discrepancy between the intermediate density $q_{k,s}(\mathbf{x})$ and the target posterior $q_{k,\infty}(\mathbf{x})$ decreases as the pseudo-time s increases. The discrepancy is measured using the Kullback–Leibler (KL) divergence,

$$\mathrm{KL}(q_{k,s} \,\|\, q_{k,\infty}) = \int q_{k,s}(\mathbf{x}) \log \frac{q_{k,s}(\mathbf{x})}{q_{k,\infty}(\mathbf{x})} \, d\mathbf{x}. \tag{84}$$

The flow field $f_{k,s}$ is assumed to lie in a reproducing kernel Hilbert space (RKHS) associated with a matrix-valued kernel

$$K : \mathbb{R}^{n_x} \times \mathbb{R}^{n_x} \to \mathbb{R}^{n_x \times n_x}. \tag{85}$$

Under the RKHS representation, the flow can be written as

$$f_{k,s}(\mathbf{x}) = \langle K(\mathbf{x}, \cdot), f_{k,s}(\cdot) \rangle_{\mathcal{H}} \in \mathbb{R}^{n_x}. \tag{86}$$

To ensure that the KL divergence decreases over pseudo-time, we require

$$\frac{d}{ds} \mathrm{KL}(q_{k,s} \,\|\, q_{k,\infty}) \leq 0. \tag{87}$$

More precisely, we aim to maximize the rate of decrease of the KL divergence, i.e., move in the steepest descent direction in function space. After calculation, the optimal flow is given by

$$f_s(\cdot) = D \int q_s(\mathbf{x}) \left[\nabla_{\mathbf{x}} \cdot K(\mathbf{x}, \cdot) + K(\mathbf{x}, \cdot) \nabla_{\mathbf{x}} \log p(\mathbf{x} \mid \mathbf{y}) \right] d\mathbf{x} = D I_f, \tag{88}$$

where D is a positive-definite matrix; for notational simplicity, the explicit dependence of f_s and q_s on the time index k is suppressed.

Algorithm 10 Particle Flow Filter Update at Time k

Input: Prior particles $\{\mathbf{x}_{k-1}^j\}_{j=1}^N$, observation $\mathbf{y}_k$, observation operator $H(\cdot)$, likelihood $p(\mathbf{y}_k \mid \mathbf{x})$, prior density $p(\mathbf{x})$
Initialization:
for $j = 1, \ldots, N$ **do**
    $\mathbf{x}_{k,0}^j \leftarrow H(\mathbf{x}_{k-1}^j, \mathbf{y}_k)$
$i \leftarrow 0$
repeat
    for $j = 1, \ldots, N_p$ **do**
        $\mathbf{x}_{k,s+1}^j \leftarrow \mathbf{x}_{k,s}^j + \varepsilon\, f_{k,s}\big(\mathbf{x}_{k,s}^j\big)$
    $s \leftarrow s + 1$
until stopping criterion is met
Output: Posterior particles $\{\mathbf{x}_{k,s}^j\}_{j=1}^N$

The derivative of the KL divergence along the flow satisfies

$$\frac{d}{ds}\mathrm{KL}(q_{k,s} \,\|\, q_{k,\infty}) = -\langle I_f, DI_f \rangle \leq 0, \tag{89}$$

which guarantees decrease since $D \succ 0$. The matrix D is chosen to ensure correct physical dimensions of the flow and is commonly set to a localized prior covariance matrix. Algorithm 10 illustrates the implementation of the Particle Flow Filter.

The implementation of each algorithm is provided in the `deterministic_kernel_flows` directory of our GitHub repository. The file `pff.py` contains the implementation of the Particle Flow Filter (PFF) algorithms. The notebook `pff_demo_100_dimension.ipynb` shows the demonstration and includes a comparison between the scalar kernel and the diagonal matrix-valued kernel. In `pff_showcase_matrix_valued_kernel.ipynb`, we conduct experiments to reproduce figures analogous to Figures 2–3 in Hu and Van Leeuwen (2021). Additional implementation details are available in the corresponding source code within the GitHub repository.

Compare EDH, LEDH, and kernel PFF on the SSM you designed in the last particle-filter question. Analyze when each method excels or fails (nonlinearity, observation sparsity, dimension, conditioning). Include stability diagnostics (flow magnitude, Jacobian conditioning).

In the notebook `compare_edh_ledh_pff_with_range_model.ipynb`, We implement the EDH and LEDH algorithms using the state-space model designed in the previous particle-filter question.

The current PFF implementation in `pff.py` does not yet generalize to arbitrary datasets; extending the implementation to more general settings is planned as future work.

Finally, we present a brief comparison of EDH, LEDH, and PFF based on the preceding experiments and the theoretical formulations of each algorithm.

Section C. Stochastic Particle Flow and Differentiable PF (Part II)

a) Due to time limitations, I was not able to implement the stochastic particle flow. As stochastic particle flow is closely related to deterministic particle flow, I hope that my implementation of deterministic particle flow reflects my understanding of the stochastic version. In addition, I have read the two papers on stochastic particle flow above.

Roughly speaking, Stochastic particle flow extends deterministic particle flow by introducing a non-zero diffusion term in the particle dynamics, allowing particles to evolve according to a stochastic process during the homotopy transformation. Deterministic particle flow is therefore a special case with zero diffusion. Dai and Daum (2021) derive a parameterized family of stochastic particle flows in which the diffusion term improves numerical stability and provides a unified framework encompassing several existing particle flow methods, including the EDH flow. Under Gaussian assumptions, they further show that the resulting particle flow estimators are unbiased and consistent.

In particle flow filters (PFFs), particles evolve according to differential equations (Crouse, 2021). However, these equations can be stiff, making numerical integration difficult. As discussed in Dai and Daum (2022), stiffness arises when the Jacobian of the flow dynamics has eigenvalues with widely different magnitudes, forcing numerical solvers to take very small integration steps. This makes it difficult to distinguish numerical errors from theoretical limitations of the flow. To address this issue, Dai and Daum (2021) show that adding a diffusion term can improve numerical stability under straight-line log-homotopy transformations. Later, Dai and Daum (2022) generalize this idea by introducing λ-varying linear log-homotopy transformations and formulating stiffness mitigation as an optimal control problem. Detailed theoretical analysis is provided in Dai and Daum (2021, 2022).

b) This is an interesting question. In the LEDH particle flow particle filter proposed by Li and Coates (2017), the deterministic particle flow is used to construct the proposal distribution. In principle, one could also consider using a stochastic particle flow as the proposal. However, I have not implemented such an approach. Intuitively, the performance may depend on the specific dataset or problem. Introducing a stochastic component into the particle flow may improve robustness and exploration of the state space, which could lead to more stable results in some cases. For example, Dai and Daum (2021, 2022) show that the inclusion of a non-zero diffusion term in stochastic particle flow can improve numerical stability, whereas deterministic particle flow has a zero diffusion term.

i) Because traditional resampling methods, such as multinomial sampling, are not differentiable, differentiable particle filters require alternative differentiable resampling schemes.

Soft resampling interpolates between the particle weight distribution and a uniform distribution Karkus et al. (2018). The modified weights are defined as $\tilde{W}_t^i = \lambda W_t^i + (1-\lambda)/N_p$, where W_t^i are the normalized weights, N_p is the number of particles, and $\lambda \in [0, 1]$ controls the interpolation. Particles are then sampled from this softened multinomial distribution,

with importance weights corrected afterward. This maintains dependence on the original weights, allowing non-zero gradients with respect to pre-resampling weights (Chen and Li, 2023; Corenflos et al., 2021).

An alternative approach formulates resampling as an optimal transport (OT) problem using entropy-regularised optimal transport (Corenflos et al., 2021). In this framework, particles are transported from an equally weighted empirical distribution to the target weighted distribution. Instead of discrete sampling, an optimal transport plan redistributes particle mass with minimal cost. New particles are obtained via barycentric projection, i.e., weighted combinations of the original particles. Entropy regularisation enables efficient and differentiable optimisation using the Sinkhorn algorithm. This results in a deterministic and differentiable mapping from the original to the resampled particles, allowing gradients to propagate through the resampling step. The entropy regularisation introduces a small bias that decreases as the regularisation coefficient approaches zero.

I have implemented these resampling methods in the `resampling/dpf/differentiable_particle_flows` folder. The code is available here).

Examples of tuning the regularisation parameter and the number of iterations for the bias–variance–speed trade-off can be found in `pf_demo_requlariseed_ot_resampling.ipynb`. An example of soft resampling is available in `pf_demo_soft_resampling.ipynb`, both located in the `examples/differentiable_particle_flows` folder.

ii) Although I have not implemented other approaches for differentiable particle filters (DPFs) yet, I can outline how I would compare them with OT resampling if I were to run experiments. To evaluate the **accuracy**, I would consider the **root mean square error** (RMSE) of the state estimates and the **log-likelihood of the observations**, since these metrics indicate how well the particle filter approximates the true posterior distribution. I would also examine the **effective sample size** (ESS) to assess particle degeneracy and the diversity of the particle set.

For **efficiency**, I would evaluate the **runtime per time step** and **memory usage**, similar to the analysis conducted in Part I. These metrics would help assess the computational cost of different DPF approaches, especially because some methods, such as OT-based resampling, can be more computationally expensive.

We consider the nonlinear state-space model described in Section 3.1 of Andrieu et al. (2010). The latent state and observation equations are

$$x_1 \sim \mathcal{N}(0, 5) \tag{90}$$

$$x_t = \frac{x_{t-1}}{2} + \frac{25x_{t-1}}{1 + x_{t-1}^2} + 8\cos(1.2t) + v_t, \quad v_t \sim \mathcal{N}(0, \sigma_v^2) \tag{91}$$

$$y_t = \frac{x_t^2}{20} + w_t, \quad w_t \sim \mathcal{N}(0, \sigma_w^2) \tag{92}$$

where the parameters are

$$\theta = (\sigma_v, \sigma_w)$$

The objective is to perform Bayesian inference for θ given observations $y_{1:T}$ using particle filtering methods within an MCMC framework.

This model is highly nonlinear and the posterior distribution can be multimodal due to the quadratic observation function.

(a) Estimation using Invertible PF-PF of Li and Coates (2017)

In this experiment the model parameters θ are assumed to be known. Therefore the objective is only to infer the latent states $x_{1:T}$ given the observations $y_{1:T}$.

Due to time constraints, the full implementation was not completed; however, the main implementation component involves computing the Jacobian determinant associated with the particle flow transformation.

We consider the invertible particle-flow particle filter (PF-PF) proposed by Li and Coates (2017). The PF-PF algorithm constructs an invertible particle flow transformation that moves particles from the predictive distribution toward regions of high likelihood given the current observation. Let the particle flow transformation be $x'_t = T(x_t)$ where $T(\cdot)$ is an invertible mapping. The proposal density is corrected using the Jacobian determinant of the transformation. This particle filter is run sequentially for $t = 1, \ldots, T$ to obtain an approximation of the filtering distribution $p(x_t|y_{1:t})$.

(b) Differentiable Particle Filters with OT Resampling and HMC

Amri et al. (2025) propose a Particle Hamiltonian Monte Carlo (PHMC) method that integrates Hamiltonian Monte Carlo within a Sequential Monte Carlo framework. Their approach addresses the limitation that HMC requires gradients of the log-target density.

Furthermore, Corenflos et al. (2021) introduce entropy-regularized optimal transport (OT) resampling, which makes the resampling step of the particle filter differentiable. This enables gradients to propagate through the particle filtering procedure and allows the use of gradient-based optimization or sampling methods.

In principle, combining differentiable particle filtering with Hamiltonian Monte Carlo could allow gradient-based inference for state-space models. Therefore, it may be possible to apply HMC together with the invertible particle-flow particle filter of Li and Coates (2017), although such a combination would require careful treatment of gradient estimation and the bias introduced by OT regularization.

(a) Using GradNetOT to Avoid Repeated Sinkhorn Iterations

Yes, the idea in Chaudhari et al. (2025) can be used to avoid repeatedly applying the Sinkhorn algorithm. Instead of solving a new optimal transport (OT) problem at every step, we can train a neural network to approximate the optimal transport map directly. The paper

proposes GradNetOT, where a monotone gradient neural network learns a transport map T_θ that pushes a source distribution p to a target distribution q, based on Brenier's theorem and the Monge–Ampère equation.

However, in many applications such as OT-based resampling, the source and target distributions **change over time**. In the paper, the neural network is trained for a *fixed pair of distributions*. If the distributions change, retraining would normally be required. To avoid repeatedly learning a new network, we can instead train a *conditional (amortized) neural network* that takes the parameters defining the current OT problem as inputs.

Instead of learning a transport map of the form

$$T_\theta(x),$$

we learn a conditional transport map

$$T_\theta(x; \mu, \nu, \lambda),$$

where the additional inputs describe the current OT problem instance.

The neural network should receive the following additional variables:

1. **Source distribution information (μ).** This describes the current source measure. For example:
   - particle locations x_i and weights a_i in a particle approximation, or
   - parameters or sufficient statistics of the source distribution (e.g., means and covariances).
2. **Target distribution information (ν).** This describes the target measure, such as:
   - particle locations y_j and weights b_j, or
   - parameters or sufficient statistics defining the target density.
3. **The transported point x.** The network must take as input the specific particle or state being transported.
4. **Model or contextual parameters.** If the source or target distributions depend on model parameters (e.g., the observation y_t, likelihood parameters, transition parameters, or time index t), these should also be included as inputs.

With this formulation, the neural network is trained *offline* over many sampled OT problems, learning a general mapping from problem parameters to the corresponding transport map. At inference time, each new OT computation only requires a *single forward pass* of the neural network, avoiding repeated Sinkhorn iterations.

In summary, by conditioning the GradNetOT network on the variables that define the source distribution, target distribution, and model parameters, we can learn a single amortized OT solver that replaces repeated Sinkhorn computations with fast neural network evaluations.

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